

MOLCHANOVO, Russian Federation

WMO: 293320

Lat: 57.576N

Lon: 83.760E

Elev: 112

StdP: 99.98

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.0	-33.2	-39.2	0.1	-35.9	-36.5	0.1	-33.2	8.5	-9.7	8.0	-11.5	1.2	290	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	28.7	18.8	27.1	18.2	25.5	17.6	20.5	26.1	19.6	25.0	18.6	23.4	2.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	23.4	17.5	12.7	22.0	16.7	12.1	21.2	59.7	26.1	56.6	25.2	53.4	23.3	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.4	5.6	-38.7	30.9	3.6	1.4	-41.3	31.9	-43.4	32.7	-45.4	33.5	-48.0	34.5
			-38.8	22.3	3.6	0.9	-41.4	23.0	-43.5	23.5	-45.5	24.0	-48.1	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-17.9	-14.4	-4.7	3.3	8.9	17.4	19.1	16.1	9.2	1.0	-10.2	-14.3	(6)
(7)		DBStd	14.13	8.20	7.82	5.54	4.78	5.43	4.85	3.28	3.47	4.75	5.26	8.10	9.09	(7)
(8)		HDD10.0	4021	865	683	455	206	88	8	0	2	72	283	607	752	(8)
(9)		HDD18.3	6370	1123	916	713	450	297	71	32	87	275	538	857	1010	(9)
(10)		CDD10.0	812	0	0	0	6	53	231	281	189	48	3	0	0	(10)
(11)		CDD18.3	120	0	0	0	0	4	43	54	17	2	0	0	0	(11)
(12)		CDH23.3	900	0	0	0	2	41	369	379	99	9	0	0	0	(12)
(13)		CDH26.7	179	0	0	0	0	7	85	76	12	0	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.9	2.8	3.3	3.2	2.9	2.2	1.9	2.0	2.4	3.0	3.0	3.0	(14)
(15)	Precipitation	PrecAvg	474	24	15	16	26	49	59	62	65	47	44	38	30	(15)
(16)		PrecMax	615	50	45	40	84	102	117	155	144	99	90	62	117	(16)
(17)		PrecMin	327	4	1	0	1	14	21	28	7	11	14	10	8	(17)
(18)		PrecStd	62	11	10	9	14	20	25	28	36	22	18	12	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.5	1.6	7.3	19.0	27.6	31.0	30.8	28.3	24.7	16.7	6.0	0.8	(19)
(20)			MCWB	-1.6	0.0	2.1	9.5	15.3	18.0	20.6	19.8	16.6	9.5	4.9	-0.2	(20)
(21)		2%	DB	-3.1	0.1	5.2	15.7	24.0	29.0	28.7	26.1	21.6	13.6	2.2	-0.3	(21)
(22)			MCWB	-3.6	-1.0	0.7	8.5	14.0	18.2	19.7	18.2	14.9	8.4	0.9	-0.9	(22)
(23)		5%	DB	-4.8	-1.5	3.8	13.1	21.2	27.3	27.1	24.1	19.0	10.5	0.9	-1.6	(23)
(24)			MCWB	-5.4	-2.5	0.7	6.9	12.5	17.6	18.8	17.3	13.2	6.9	0.2	-2.2	(24)
(25)		10%	DB	-6.5	-3.8	2.4	10.7	17.9	25.4	25.5	22.3	16.6	8.2	-0.2	-3.4	(25)
(26)			MCWB	-7.0	-4.5	0.1	5.4	10.9	17.2	18.4	16.5	12.2	5.3	-0.9	-4.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	0.6	3.4	11.1	16.0	21.3	22.5	20.7	17.2	10.3	5.3	0.1	(27)
(28)			MCDWB	-0.9	1.2	5.7	17.4	23.9	27.1	28.2	26.5	23.3	14.9	6.2	0.6	(28)
(29)		2%	WB	-3.5	-0.8	2.0	9.4	14.1	20.0	21.0	19.3	15.3	8.7	1.2	-0.8	(29)
(30)			MCDWB	-3.1	0.1	3.6	14.4	21.7	25.9	26.6	23.9	20.6	12.6	1.8	-0.2	(30)
(31)		5%	WB	-5.5	-2.5	1.0	7.4	12.8	19.0	20.0	18.3	13.9	7.2	0.2	-2.1	(31)
(32)			MCDWB	-5.0	-1.5	2.9	12.4	19.8	25.1	25.6	22.2	17.9	10.2	0.8	-1.6	(32)
(33)		10%	WB	-7.0	-4.8	0.2	5.7	11.4	17.8	19.1	17.4	12.4	5.6	-0.8	-4.0	(33)
(34)			MCDWB	-6.5	-3.9	2.5	10.2	17.3	23.5	24.1	21.0	16.0	8.1	-0.1	-3.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.2	7.7	8.9	9.1	9.8	9.2	8.7	7.8	5.6	5.9	6.0	(35)
(36)			MCDBR	7.9	6.0	8.2	11.9	12.6	12.3	11.5	11.2	11.5	9.2	4.6	6.8	(36)
(37)			MCWBR	7.7	5.5	5.6	7.0	6.0	4.6	4.5	5.2	5.7	5.9	4.5	6.7	(37)
(38)		5% WB	MCDBR	7.9	5.9	6.6	11.1	12.1	10.7	10.2	9.3	10.9	9.1	4.6	6.7	(38)
(39)			MCWBR	7.8	5.5	5.0	7.0	6.8	4.7	4.9	4.7	5.8	6.2	4.6	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.281	0.314	0.361	0.388	0.398	0.402	0.438	0.404	0.364	0.340	0.303	0.281	(40)	
(41)		taud	2.000	1.990	1.941	2.086	2.258	2.326	2.210	2.294	2.358	2.263	2.044	1.942	(41)	
(42)		Ebn at Noon	531	687	756	813	836	837	792	795	774	672	495	394	(42)	
(43)		Edh at Noon	60	96	133	136	123	117	129	110	89	74	57	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.68	1.66	3.11	4.44	4.93	5.70	5.22	4.08	2.62	1.32	0.61	0.39	(44)	
(45)		RadStd	0.08	0.13	0.22	0.35	0.48	0.60	0.43	0.35	0.36	0.12	0.11	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page